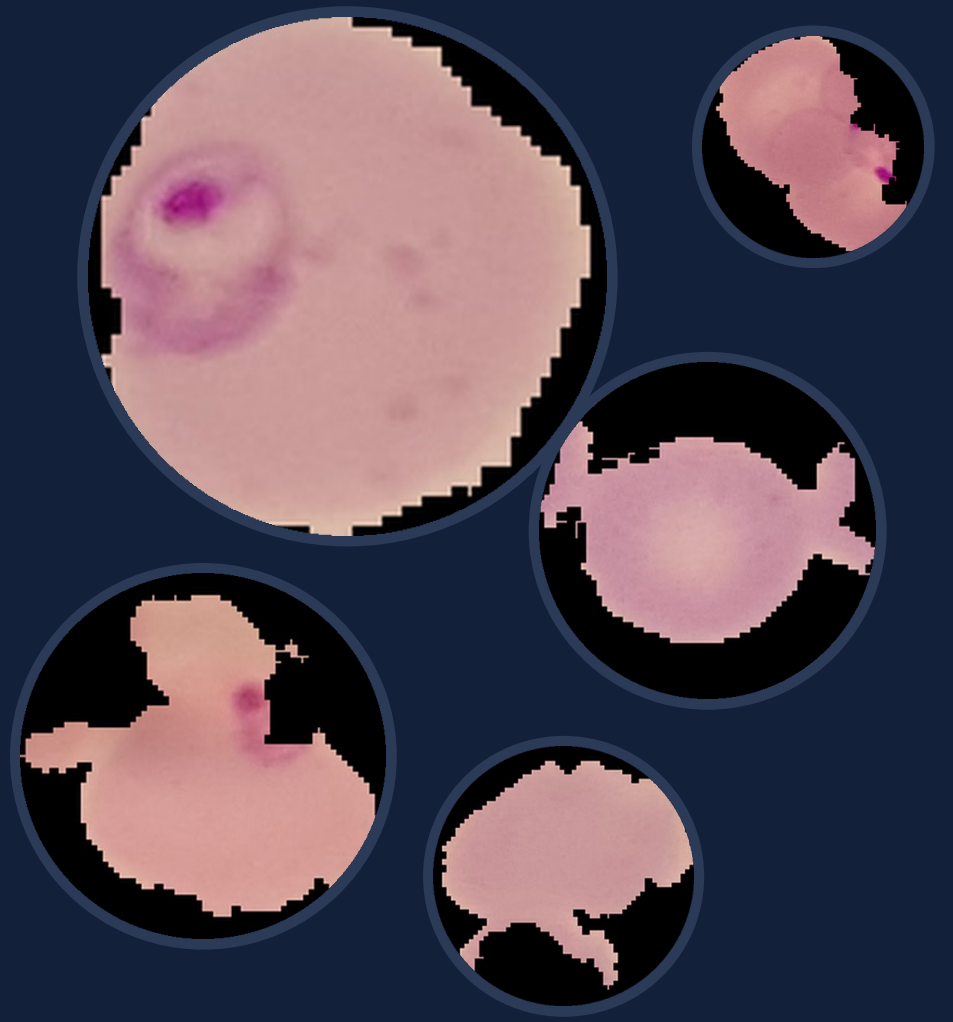
CV RESEARCH-REPRODUCTION PROJECT · FINAL DEFENCE

98 % Accuracy Holds on Unseen Patients



Code-exact reproduction and pre-registered patient-level validation of an EfficientNet-B0 malaria ensemble

J Amathziah
230139

C Murali Madhav
230115

Ravi Yadav
230131

Pugazhendhi J
230066

Department of Computer Science · Rishihood University, Sonipat · github.com/amathziah/malaria-detection

AT A GLANCE

What we achieved

Within 1 pp

of the paper on every metric, rebuilt from the authors' own code

25× faster

11 minutes per fold instead of 4 h 46 min, with the same maths

98.45 %

accuracy on 40 unseen patients; recall 98.05 %, ROC-AUC 99.80 %

0.85 pp

patient leakage in MCC, measured on 17,591 identical cells: small

±1 pp

YUV stain normalisation is unnecessary: equivalent to no change

26 of 28

infected patients found and 12 of 12 healthy cleared (exploratory)

Pre-registered before training · patient-level statistics · all 40 models and the code are public

ROADMAP

Five parts, following the brief's six steps



PART 1

The problem

27,558 cells · 200 patients

Brief step 1 · slides 4 to 6

PART 2

Reproduction

within 1 pp · 25× faster

Brief steps 2 and 3 · slides 7 to 9

PART 3

Our change

2 predictions, fixed before training

Brief step 4 · slide 10

PART 4

Results

98.45 % on unseen patients

Brief step 5 · slides 11 to 14

PART 5

Failure analysis and conclusion

26 of 28 patients found

Brief step 6 · slides 15 to 17

The tracker at the top right of every slide shows where we are.

Automating malaria microscopy, one cell at a time

Slow

Hundreds of cells per smear, checked by eye

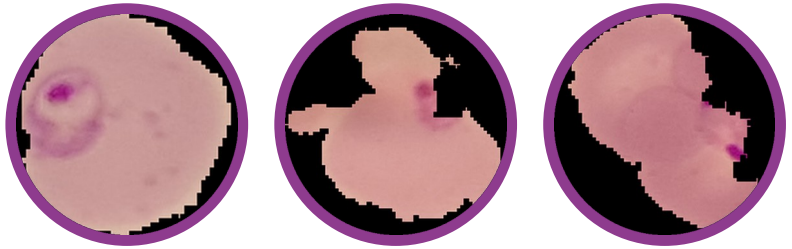
Expert-dependent

Needs trained microscopists, scarce where malaria is common

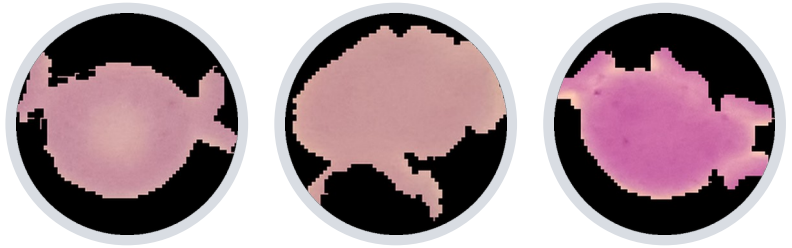
Variable

Calls differ between observers and laboratories

Our goal call each red blood cell parasitized or uninfected, accurately on new patients.



parasitized parasitized parasitized



uninfected uninfected uninfected

27,558

single-cell images from the NIH dataset, half parasitized

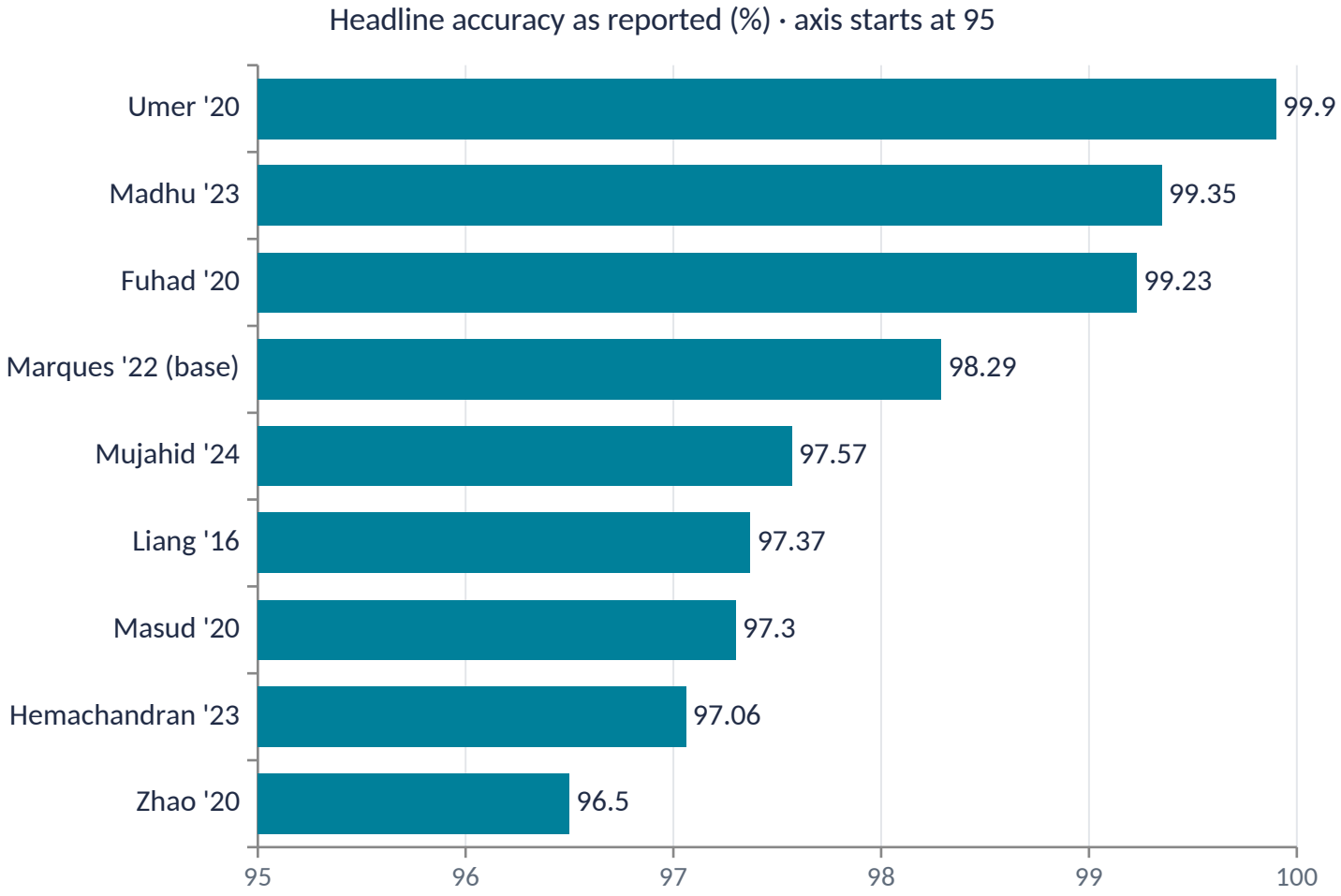
200

patients: 150 infected, 50 healthy

`C33P1thinF_IMG_…_cell_179.png`

Every file name starts with the patient ID; we verified all 27,558 against NIH's mapping.

11 studies report up to 99.9 %; 2 test unseen patients



The open question

9 of 11 split cells at random, so the same patients sit in training and test.

Hard to compare

5-fold, 10-fold and 80/20 protocols; original or cleaned labels.

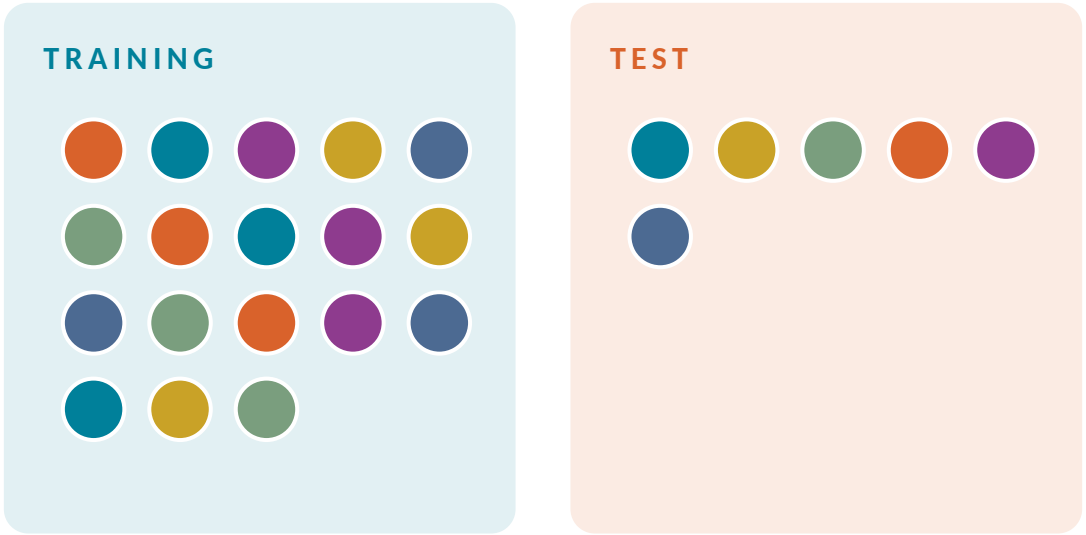
Our base paper

Marques et al. published executed code, so it can be rebuilt exactly.

Our question: does 98 % hold on new patients?

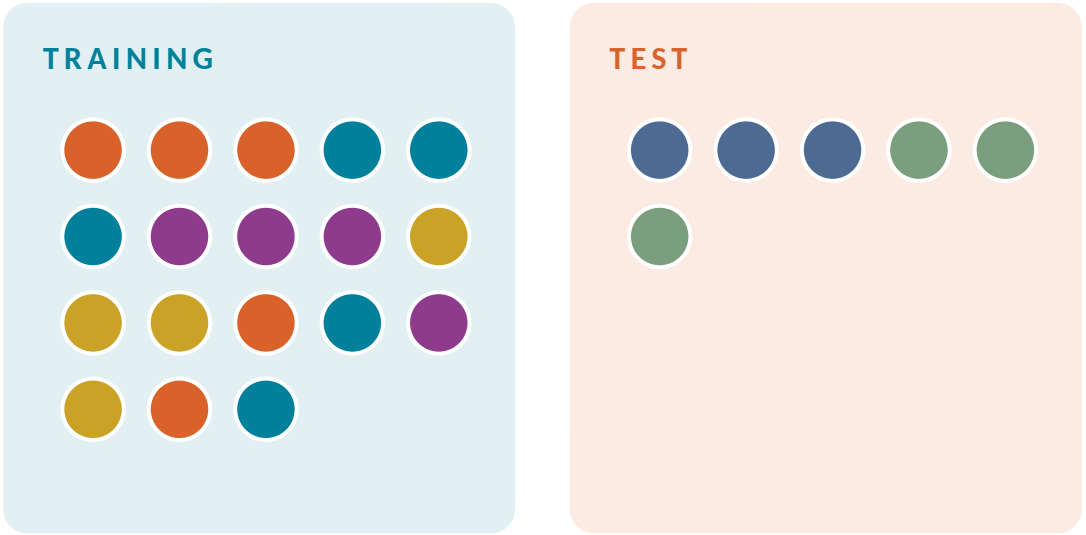
Image split (the paper)

cells shuffled at random: every patient on both sides



Patient split (ours)

whole patients go to one side only



Each colour is one patient. A patient's cells share stain, lighting and blood, so we hold out whole patients: the model must recognise the parasite, not the patient.

Base paper checklist: every gap closed

Checklist item	Paper	What we did
Data public	**Yes**	Downloaded; patient IDs verified against NIH's mapping
Code public	**Yes**	Rebuilt from the authors' executed notebook
Model size	**Yes**	4,223,934 parameters, matched exactly
Hyperparameters	**Partly**	Text and code differ in 8 places: we follow the code
Data split	**No**	Seeded and fingerprinted: identical on every machine
Evaluation protocol	**No**	Both reported: clean and augmented images
Patient-level test	**No**	Our hypothesis H1
Trained models	**No**	All 40 of ours public on GitHub

Marques et al. (2022)

8

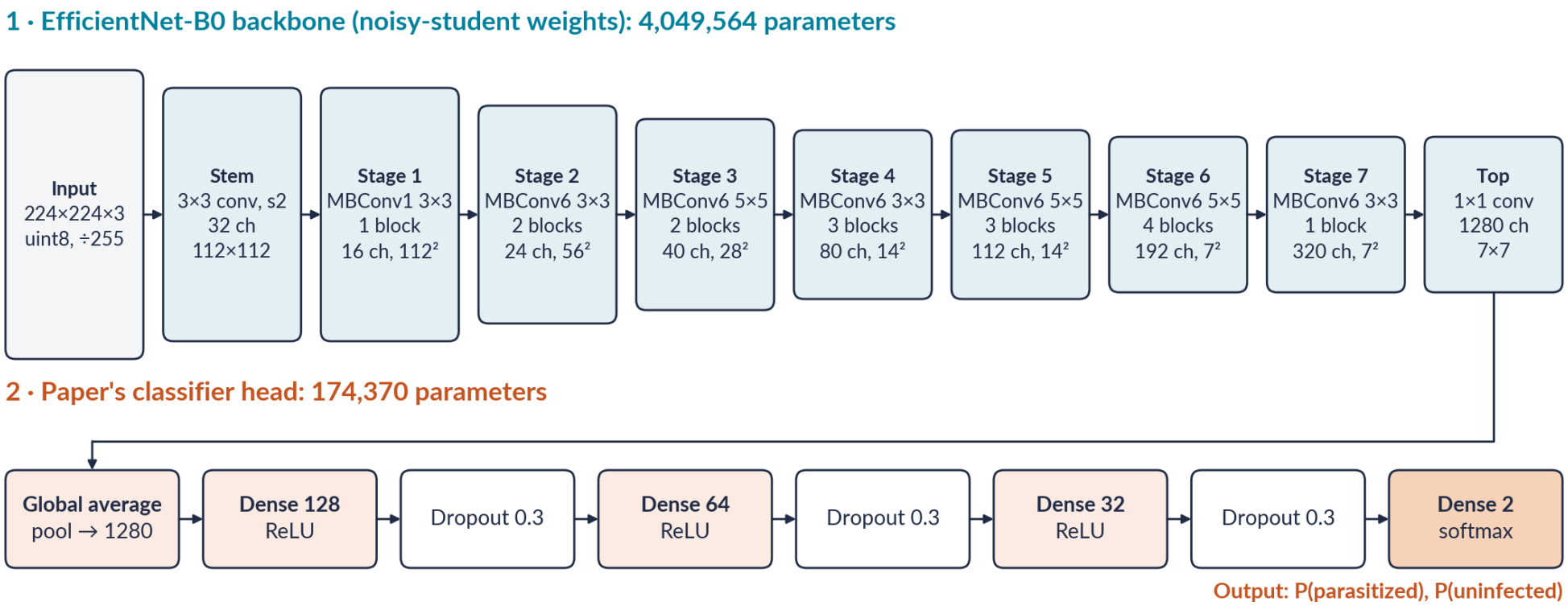
places where the code differs from the paper's text; we follow the code

3

reporting errors we found and corrected

Precision and recall swapped · test images augmented · "accuracy" = (P + R) / 2

EfficientNet-B0 and the paper's head: 4.2 M parameters

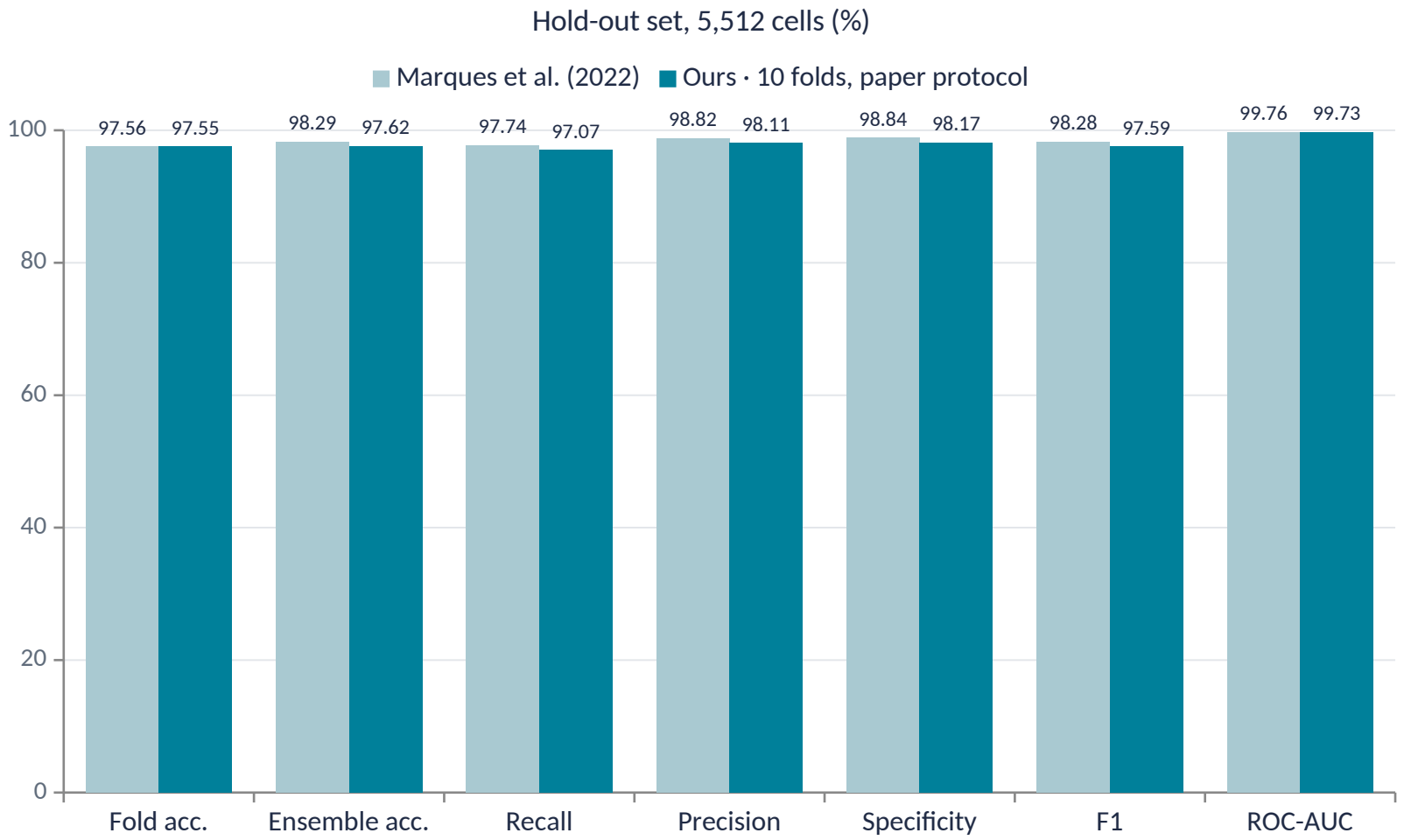


Box height shows the resolution (224 → 112 → 56 → 28 → 14 → 7 pixels) as the channels grow 3 → 32 → … → 320 → 1280.
16 MBConv blocks in 7 stages · 81 convolution layers (65 normal + 16 depthwise) · 49 batch-norm layers
230 Keras layers in the backbone · 4 dense layers in the head

Total: 4,223,934 parameters (4,181,918 trainable + 42,016 non-trainable batch-norm statistics), exactly the paper's Table 2.

Ten fold models, each trained on a different nine tenths of the data, averaged into one ensemble.

Reproduced within 1 pp of the paper, 25× faster



32

automated tests check the rebuild against the paper's outputs

97.55 %

mean fold accuracy, against the paper's 97.56 %

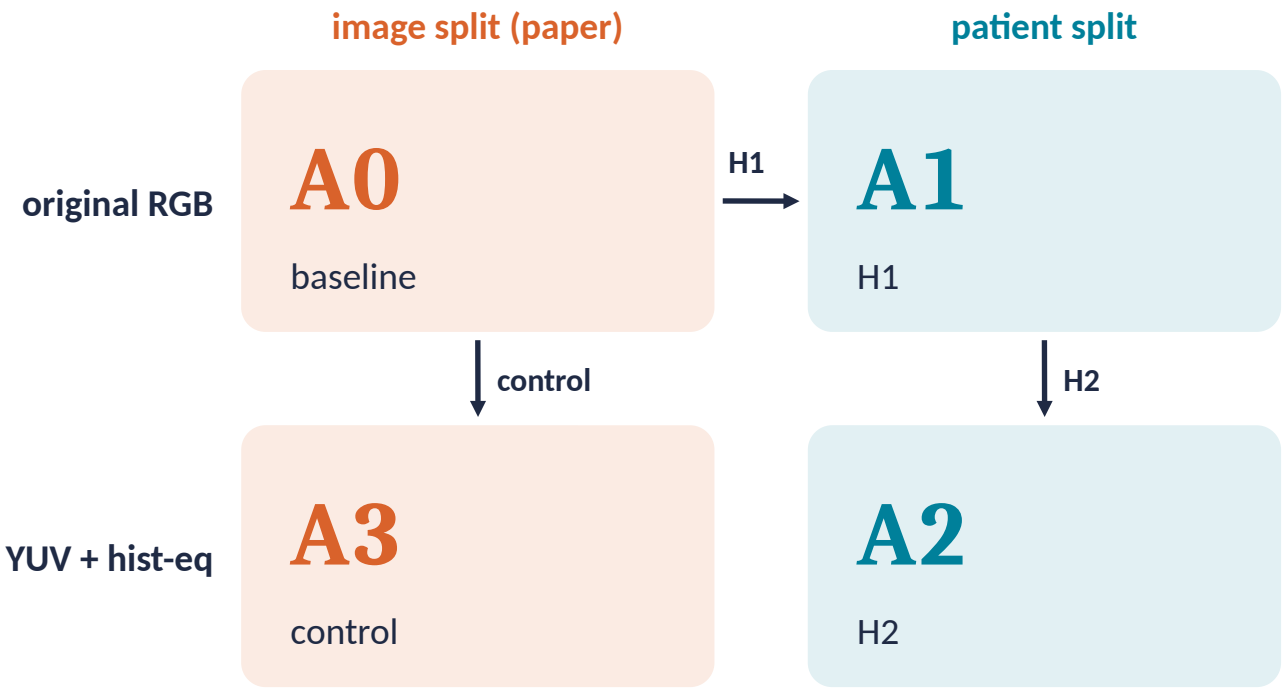
11 min

per fold on one RTX 4090, against 4 h 46 min

US$6.95

for 30 more models on ten cloud GPUs at once

Two predictions, fixed before training



H1 · PATIENT SPLIT

We predicted on unseen patients accuracy drops, and the fold-to-fold spread widens.

H2 · YUV STAIN NORMALISATION

We predicted equalising brightness raises recall and MCC on unseen patients.

Committed to GitHub before training (0c086ee): patient bootstrap · paired tests · Holm correction · MCC as the primary metric

Recall above 97 % and AUC above 99.6 % in every experiment

	A0 · image split	A1 · patient split	A2 · patient + YUV	A3 · image + YUV
Hold-out set	5,512 cells 200 patients	5,550 cells 40 patients	5,550 cells 40 patients	5,512 cells 200 patients
Ensemble accuracy	97.64 %	98.45 %	98.32 %	97.64 %
Recall (sensitivity)	**97.18 %**	**98.05 %**	**98.22 %**	**97.14 %**
Specificity	98.09 %	98.87 %	98.43 %	98.13 %
MCC × 100	95.29	96.90	96.65	95.29
ROC-AUC	**99.68 %**	**99.80 %**	**99.76 %**	**99.66 %**
Fold MCC, mean ± SD	95.57 ± 0.60	93.92 ± 1.51	93.32 ± 1.66	95.20 ± 0.37

Ten-model ensembles, clean evaluation. A1 and A2: 40 unseen patients; A0 and A3: the paper's split. 95 % intervals: paper, Table III.

98 % accuracy holds on unseen patients

98.45 %

accuracy on 40 unseen patients (95 % CI 97.94 to 98.91)

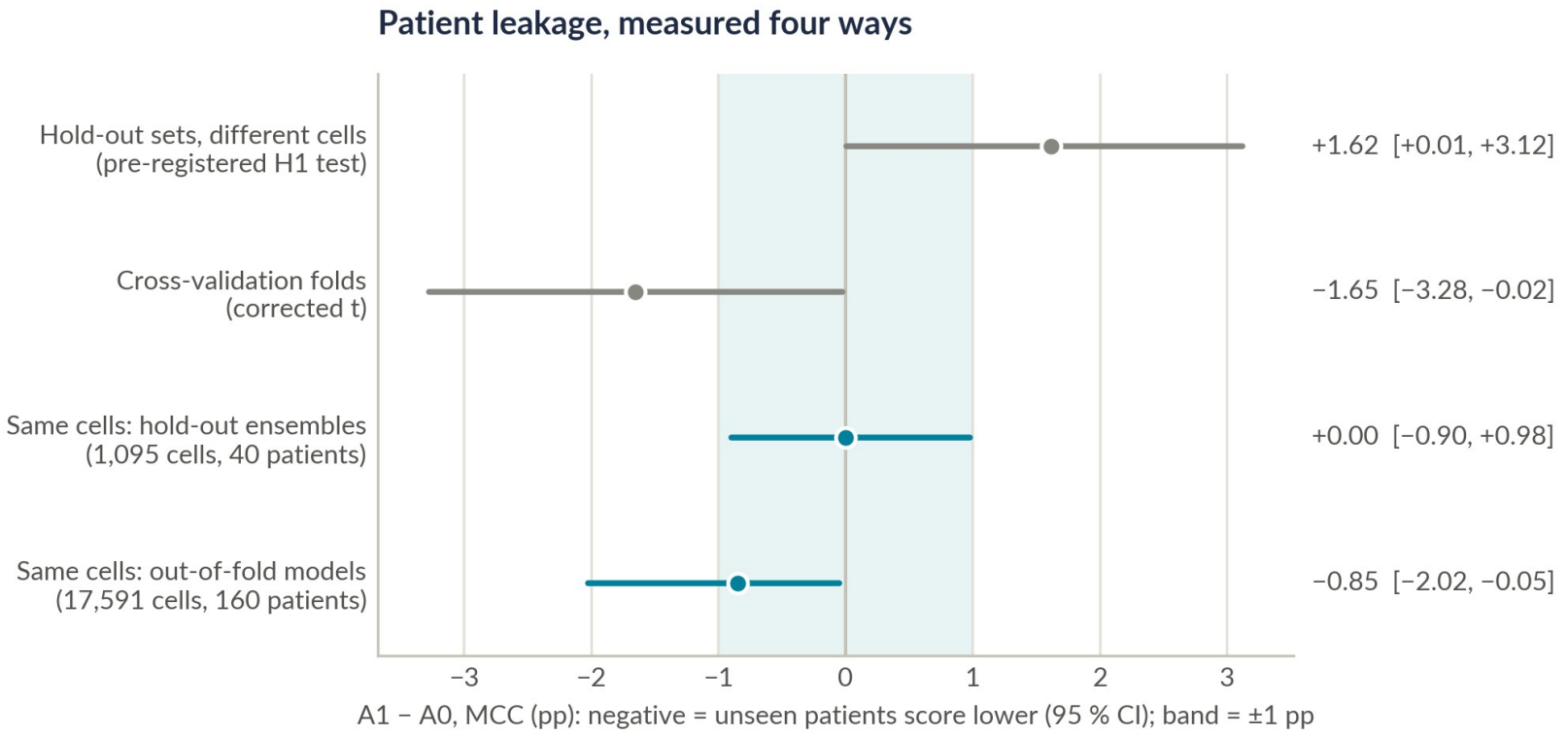
98.05 %

recall, with ROC-AUC 99.80 %

0.85 pp

patient leakage in MCC on 17,591 identical cells: small

Teal rows: both models score the same cells.



The model is already stain-robust

±1 pp

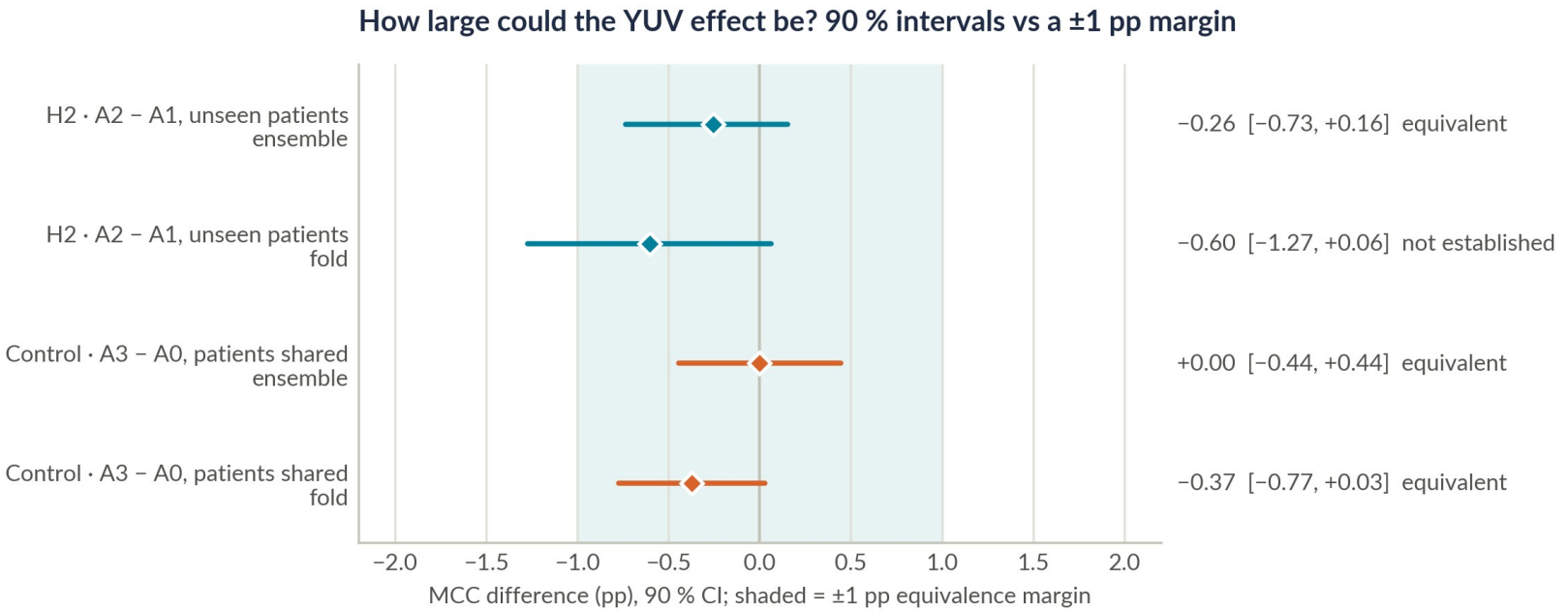
YUV's effect on MCC: 90 % CI −0.73 to +0.16

p = 0.008

equivalence test (TOST): the effect lies inside the margin

Simpler

pipeline, same accuracy: no stain normalisation step needed



How to read it: a bar wholly inside the shaded ±1 pp band means the effect is too small to matter. On the hold-out set, YUV's bar sits inside it.

What our predictions taught us

H1 · PATIENT LEAKAGE

Predicted a clear drop on unseen patients, with a wider fold spread.

Found the spread widened 2.5× as predicted; leakage is just 0.85 pp MCC.

Insight parasite chromatin looks alike in every patient, and the augmentation hides each patient's stain: the model generalises.

H2 · YUV STAIN NORMALISATION

Predicted higher recall and MCC on unseen patients.

Found equivalent to no change, within ±1 pp.

Insight the augmentation already varies brightness and contrast, so the baseline is stain-robust; the large published gain came from a baseline that had not trained.

Next test: colour augmentation, on another laboratory's smears [16].

The remaining errors point to the labels

7 of 40

patients hold half of the 86 errors

14 of 40

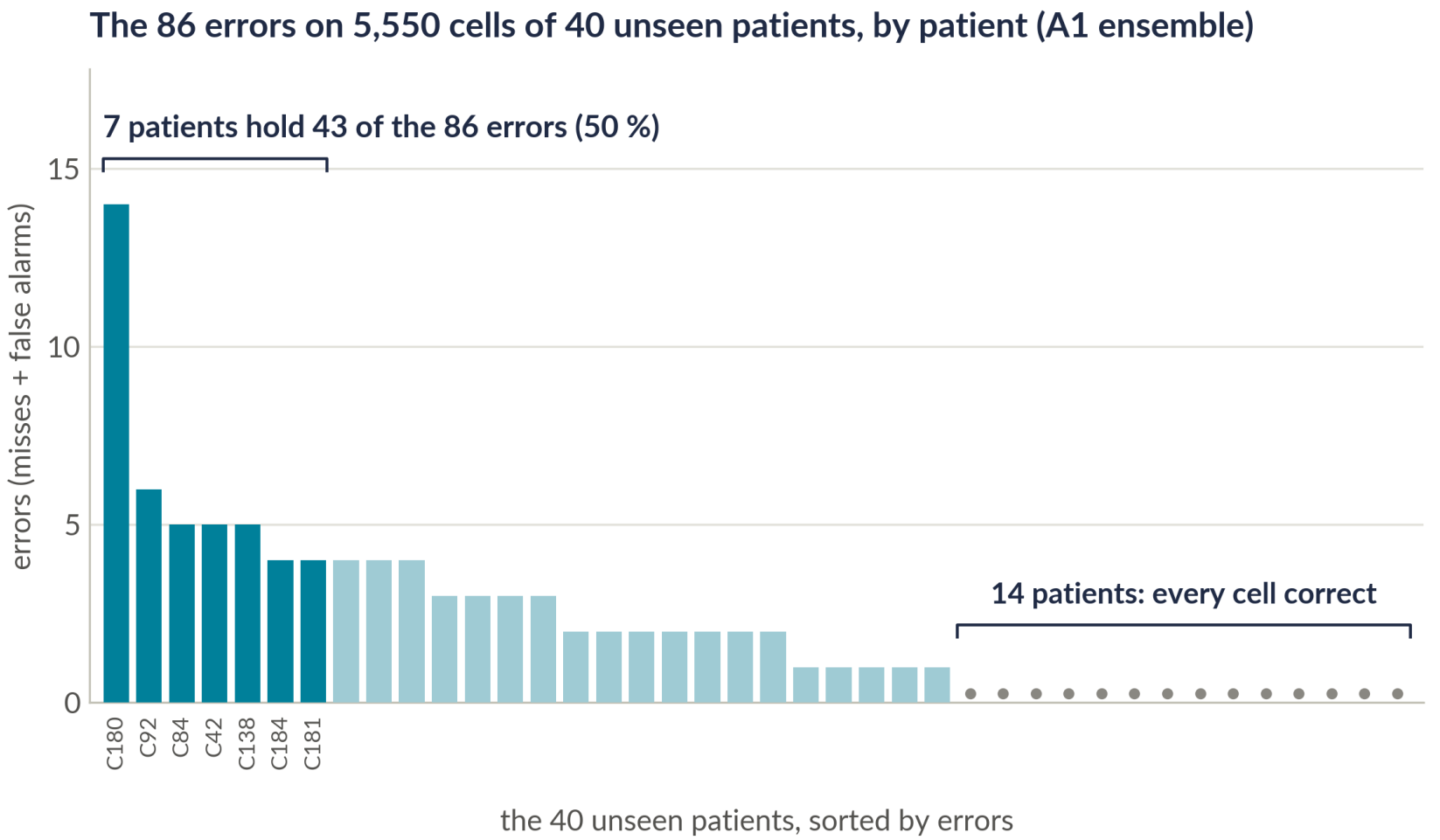
patients: every cell classified correctly

30 of 31

false alarms on infected patients' smears (1.57 % vs 0.12 %)

8

cells flagged as likely label errors (confident learning)



26 of 28 infected patients found, 12 of 12 cleared

26 / 28

infected patients found (92.9 %)

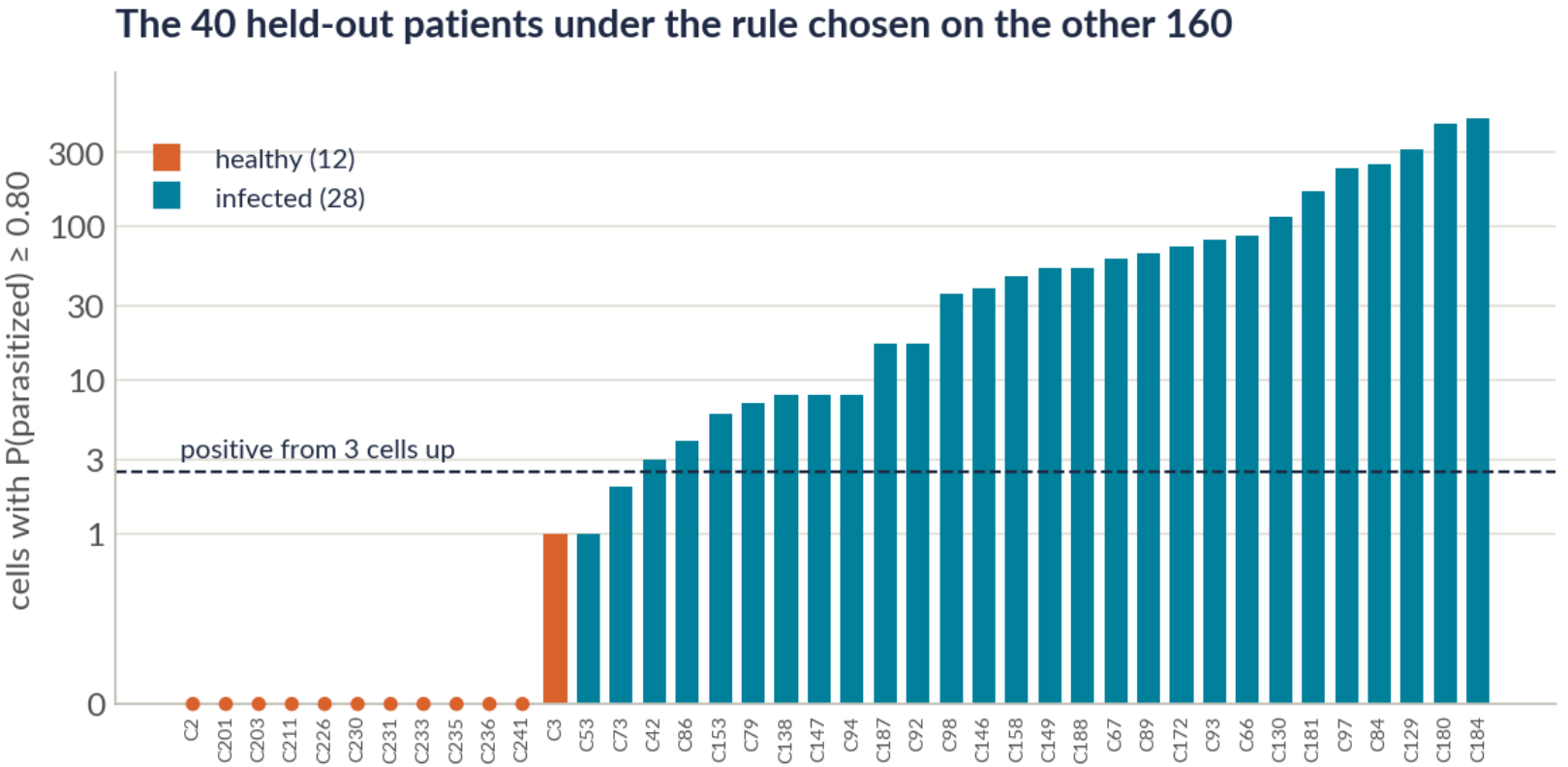
12 / 12

healthy patients cleared (100 %)

99.22 %

cell recall with a threshold chosen on the CV folds (from 98.05 %)

Rule fixed on the 160 training patients: at least 3 cells with P ≥ 0.80. Exploratory, on curated crops. Screening view, no clinical claim.



What we contribute

1

Code-exact reproduction

Within 1 pp of the paper, 25× faster, checked by 32 tests, 3 reporting errors corrected.

2

Validated on unseen patients

98.45 % accuracy and 99.80 % AUC on 40 new patients; leakage measured at 0.85 pp MCC.

3

A simpler pipeline

YUV stain normalisation is unnecessary: equivalent within ±1 pp.

4

Open and pre-registered

Plan committed before training; code and all 40 models public.

Scope

One hospital · curated cell crops · one expert's labels

Next

- Another lab's smears (MP-IDB)
- Colour augmentation
- Patient rule on whole smears
- Distil into one light model

Thank you · questions welcome · github.com/amathziah/malaria-detection

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